ROBERT v 1.2.1 2025/03/19 19:30:06

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Section A. ROBERT Score

This score is designed to evaluate the models using different metrics.

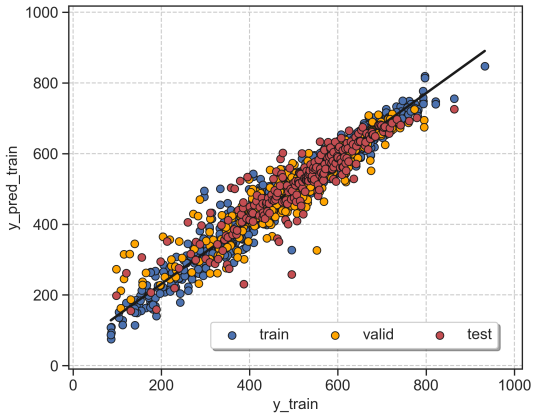
No PFI (standard descriptor filter):

Model = GB · Train:Validation:Test = 72:18:10

Points(train+valid.):descriptors = 2119:16

Score = 7 / 10

MODERATE



Train : R^2 = 0.95, MAE = 2.1e+01, RMSE = 2.9e+01

Valid. : R^2 = 0.88, MAE = 3.4e+01, RMSE = 4.8e+01

Test : R^2 = 0.84, MAE = 3.8e+01, RMSE = 5.3e+01

Severe warnings

- No severe warnings detected

Moderate warnings

- Uneven y distribution (Section C)
- Potential "faulty" outliers (Section E)

Overall assessment

- Decent model, but it has limitations

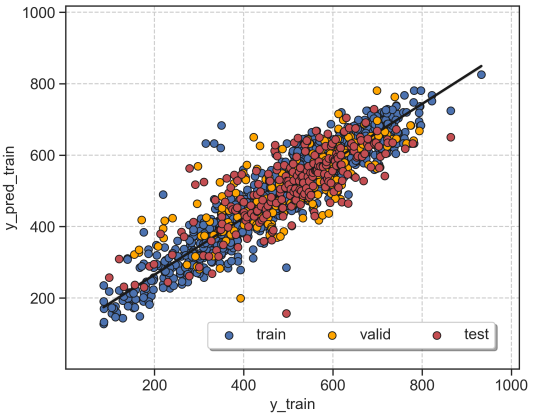
PFI (only most important descriptors):

Model = RF · Train:Validation:Test = 81:9:10

Points(train+valid.):descriptors = 2119:12

Score = 4 / 10

WEAK



Train : R^2 = 0.89, MAE = 3.3e+01, RMSE = 4.6e+01

Valid. : R^2 = 0.6, MAE = 6.3e+01, RMSE = 8.2e+01

Test : R^2 = 0.63, MAE = 6e+01, RMSE = 8e+01

Severe warnings

- No severe warnings detected

Moderate warnings

- Some tests are unclear (Section B.1)
- Uneven y distribution (Section C)
- Potential "faulty" outliers (Section E)

Overall assessment

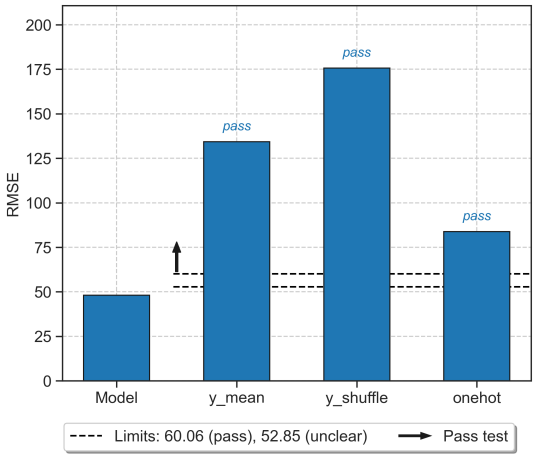
- The model is unreliable

Section B. Advanced Score Analysis

This section explains each component that comprises the ROBERT score.

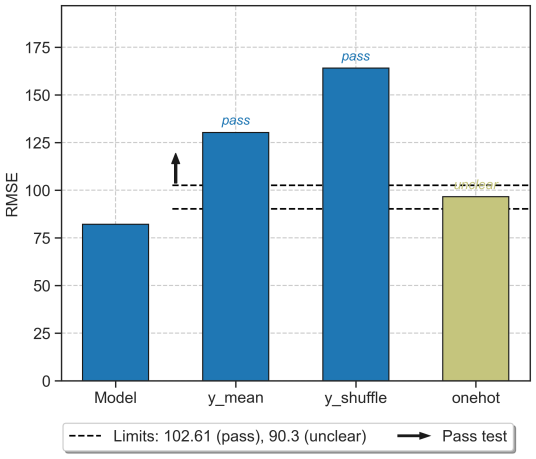
1. Model vs "flawed" models (3 / 3)

The model predicts right for the right reasons.
Pass: +1, Unclear: 0, Fail: -1. *Details here.*



1. Model vs "flawed" models (2 / 3)

WARNING! The model might have important flaws.
Pass: +1, Unclear: 0, Fail: -1. *Details here.*



2. Predictive ability of the model (1 / 2)

Moderate predictive ability with R^2 (test) = 0.84.
R^2 0.70-0.85: +1, R^2 >0.85: +2.

2. Predictive ability of the model (0 / 2)

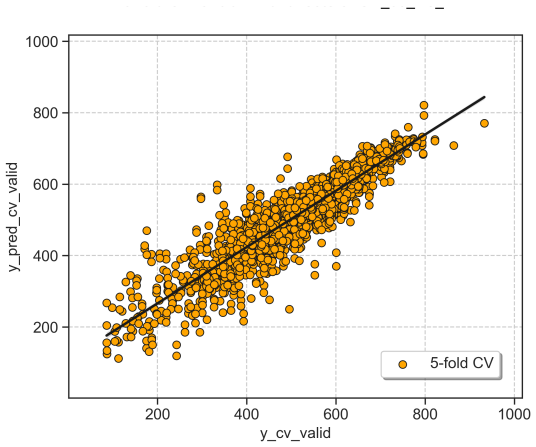
Low predictive ability with R^2 (test) = 0.63.
R^2 0.70-0.85: +1, R^2 >0.85: +2.

3. Cross-validation (5-fold CV) of the model

Overfitting analysis on the model with 3a and 3b:

3a. CV predictions train + valid. (1 / 2)

Moderate predictive ability with R^2 (5-fold CV) = 0.84.
R^2 0.70-0.85: +1, R^2 >0.85: +2.

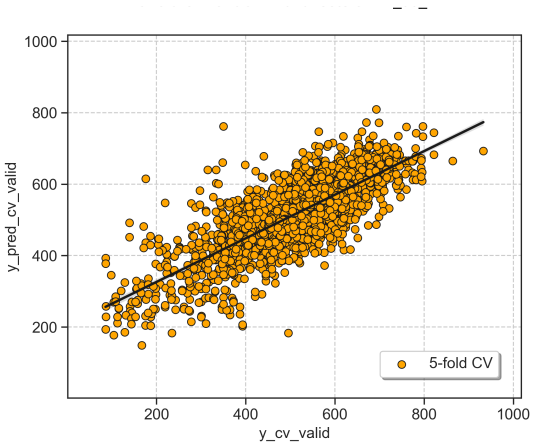


3. Cross-validation (5-fold CV) of the model

Overfitting analysis on the model with 3a and 3b:

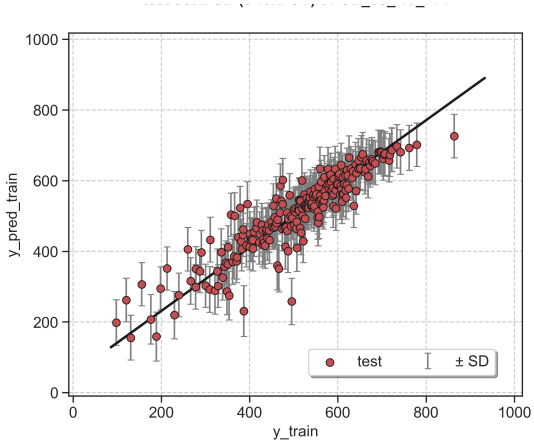
3a. CV predictions train + valid. (0 / 2)

Low predictive ability with R^2 (5-fold CV) = 0.62.
R^2 0.70-0.85: +1, R^2 >0.85: +2.



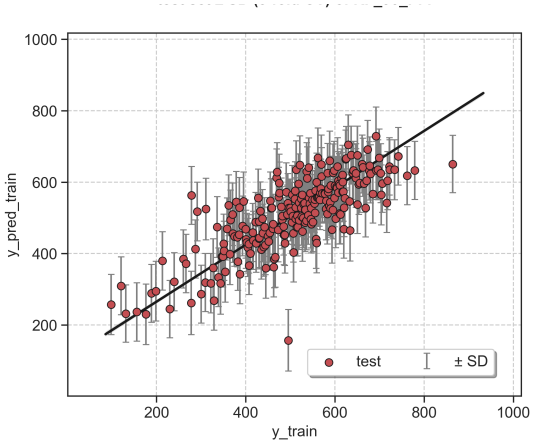
3b. Avg. standard deviation (SD) (1 / 2)

Moderate variation, 4*SD (test) = 248.9 (29% y-range).
4*SD 25-50% y-range: +1, 4*SD < 25% y-range: +2.
Details here.



3b. Avg. standard deviation (SD) (1 / 2)

Moderate variation, 4*SD (test) = 329.7 (39% y-range).
4*SD 25-50% y-range: +1, 4*SD < 25% y-range: +2.
Details here.



4. Points(train+valid.):descriptors (1 / 1)

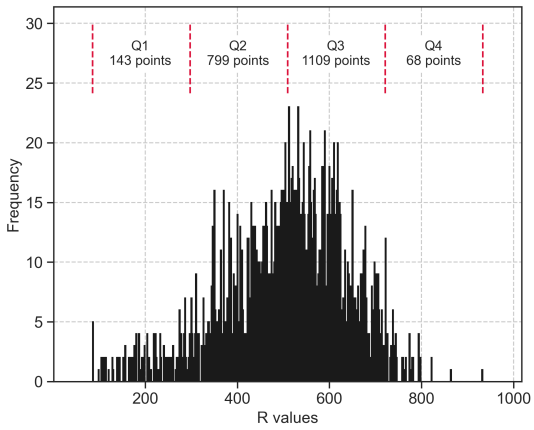
Decent number of descps. (ratio 2119:16).
5 or more points per descriptor: +1.

4. Points(train+valid.):descriptors (1 / 1)

Decent number of descps. (ratio 2119:12).
5 or more points per descriptor: +1.

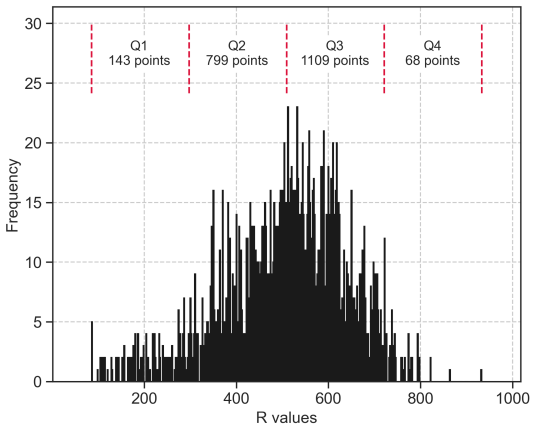
Section C. Distribution of y Values

This section shows the distribution of y values within the training and validation sets.



y distribution analysis

x WARNING! Your data is not uniform (Q4 has 68 points while Q3 has 1109)

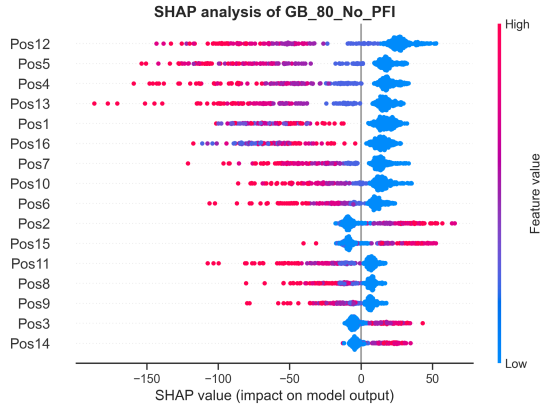


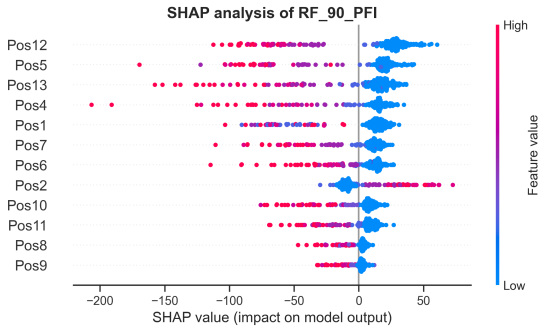
y distribution analysis

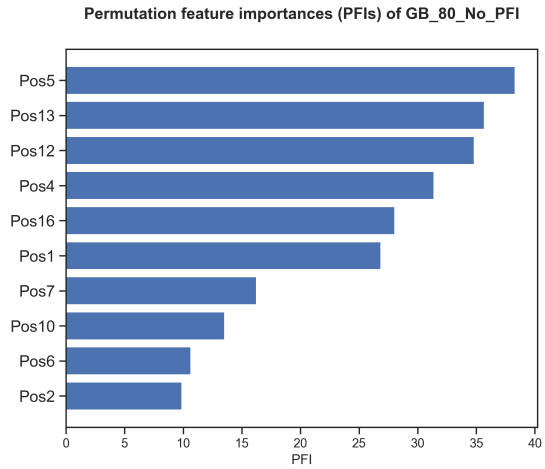
x WARNING! Your data is not uniform (Q4 has 68 points while Q3 has 1109)

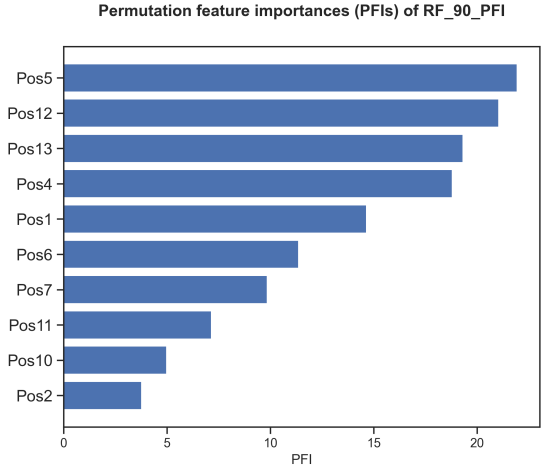
Section D. Feature Importances

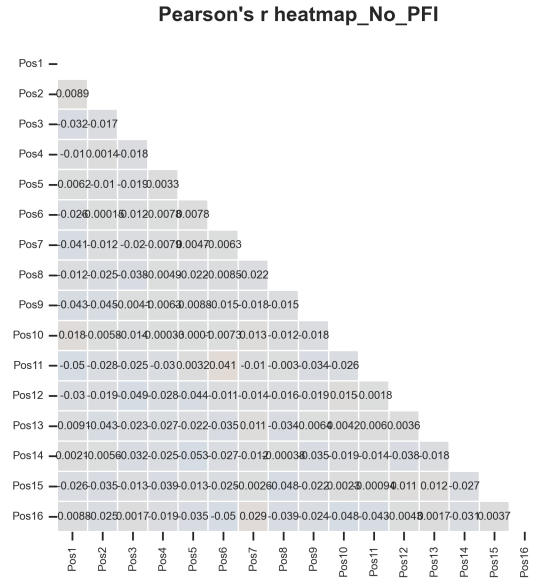
This section presents feature importances measured using the validation set.

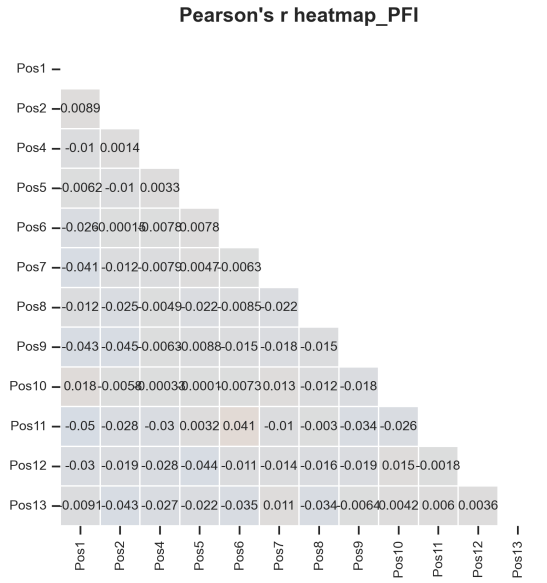












Correlation analysis

o Correlations between variables are acceptable

Correlation analysis

o Correlations between variables are acceptable

Section E. Outlier Analysis

This section detects outliers using the standard deviation (SD) of errors from the training set.

No PFI (standard descriptor filter):

Outliers (max. 10 shown)
Train: 66 outliers out of 1695 datapoints (3.9%)
- 6Heliceno_2-brom_3-brom_5-fluor_12-fluor_14-brom_15-brom.log (3.3 SDs)
- 6Heliceno_2-fluor_11-chlor_12-chlor_13-iod_15-brom.log (2.1 SDs)
- 6Heliceno_3-fluor_4-iod_11-fluor_15-fluor.log (2.1 SDs)
- 6Heliceno_5-chlor_6-iod_12-fluor_13-iod_15-fluor.log (2.2 SDs)
- 6Heliceno_2-fluor_6-iod_7-chlor_8-iod_12-chlor.log (3.1 SDs)
- 6Heliceno_3-chlor_5-chlor_8-fluor_11-iod_13-fluor_16-iod.log (2.9 SDs)
- 6Heliceno_2-chlor_4-chlor_7-fluor_12-chlor.log (5.4 SDs)
- 6Heliceno_2-iod_15-iod_16-iod.log (3.0 SDs)
- 6Heliceno_3-iod_5-fluor_14-iod.log (4.4 SDs)
- 6Heliceno_1-chlor_2-iod_3-iod.log (7.5 SDs)

Validation: 65 outliers out of 424 datapoints (15.3%)
- 6Heliceno_8-brom_11-brom_12-brom_13-iod_14-chlor.log (5.6 SDs)
- 6Heliceno_3-chlor_10-chlor_11-iod.log (2.8 SDs)
- 6Heliceno_2-brom_3-iod_6-brom.log (2.6 SDs)
- 6Heliceno_1-fluor_5-chlor_13-iod_14-iod_16-chlor.log (4.6 SDs)
- 6Heliceno_4-iod_5-iod_11-chlor_16-chlor.log (3.0 SDs)
- 6Heliceno_3-iod_5-iod_7-brom_9-iod_10-fluor.log (2.6 SDs)
- 6Heliceno_6-fluor_7-chlor_8-brom_10-fluor_12-iod_15-brom.log (3.1 SDs)
- 6Heliceno_2-chlor_3-chlor_5-chlor_14-chlor_15-chlor.log (4.1 SDs)
- 6Heliceno_2-fluor_3-chlor_4-fluor_9-chlor_13-chlor.log (3.3 SDs)
- 6Heliceno_1-fluor_3-iod_5-iod_6-fluor_15-iod_16-iod.log (7.2 SDs)

Test: 49 outliers out of 235 datapoints (20.9%)
- 6Heliceno_1-fluor_4-brom_5-brom_9-iod_11-fluor_16-brom.log (4.0 SDs)
- 6Heliceno_2-fluor_5-fluor_8-chlor_12-iod_13-brom_16-fluor.log (6.1 SDs)
- 6Heliceno_4-brom_6-fluor_8-brom_12-iod_13-iod_15-fluor.log (6.6 SDs)
- 6Heliceno_4-iod_5-chlor_8-fluor_9-iod_13-iod.log (3.8 SDs)
- 6Heliceno_4-fluor_5-fluor_6-iod_7-iod_10-brom_13-fluor.log (6.0 SDs)
- 6Heliceno_10-iod_11-iod_14-iod.log (6.4 SDs)
- 6Heliceno_6-fluor_8-chlor_11-chlor_13-fluor_15-iod_16-iod.log (2.6 SDs)
- 6Heliceno_1-brom_3-fluor_10-iod_11-iod_14-chlor.log

(4.3 SDs)
- 6Heliceno_8-iod_9-iod_10-fluor_13-iod.log (5.1 SDs)

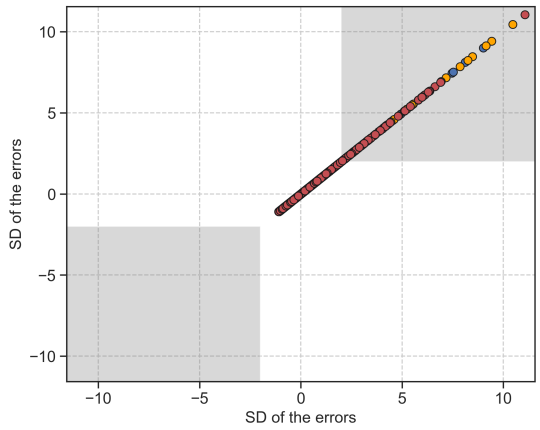
- 6Heliceno_1-iod_2-chlor_3-iod_4-chlor_12-chlor.log (2.1 SDs)

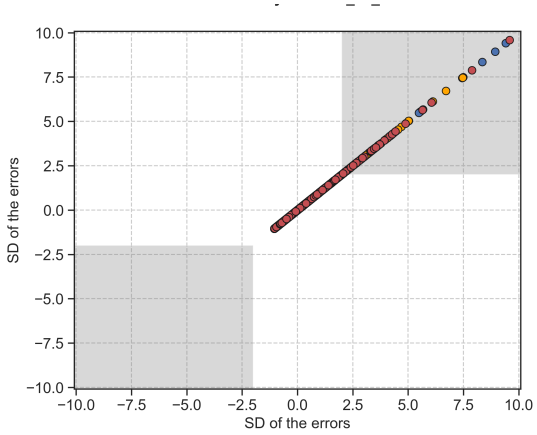
PFI (only most important descriptors):

Outliers (max. 10 shown)
Train: 81 outliers out of 1907 datapoints (4.2%)
- 6Heliceno_14-chlor_15-iod.log (3.4 SDs)
- 6Heliceno_2-brom_3-brom_5-fluor_12-fluor_14-brom_15-brom.log (2.3 SDs)
- 6Heliceno_12-iod_14-chlor_16-brom.log (3.2 SDs)
- 6Heliceno_5-chlor_13-chlor.log (2.3 SDs)
- 6Heliceno_2-fluor_6-iod_7-chlor_8-iod_12-chlor.log (2.5 SDs)
- 6Heliceno_2-iod_15-iod_16-iod.log (9.4 SDs)
- 6Heliceno_3-iod_5-fluor_14-iod.log (3.3 SDs)
- 6Heliceno_5-fluor_11-iod_12-iod_13-fluor_14-fluor_16-brom.log (2.7 SDs)
- 6Heliceno_1-chlor_2-iod_3-iod.log (3.4 SDs)
- 6Heliceno_1-iod_5-brom_7-chlor_8-chlor_14-fluor_15-brom.log (2.1 SDs)

Validation: 49 outliers out of 212 datapoints (23.1%)
- 6Heliceno_3-fluor_7-brom_14-chlor.log (2.3 SDs)
- 6Heliceno_4-fluor_16-chlor.log (2.1 SDs)
- 6Heliceno_1-fluor_5-chlor_13-iod_14-iod_16-chlor.log (4.7 SDs)
- 6Heliceno_2-chlor_13-iod.log (3.4 SDs)
- 6Heliceno_4-fluor_6-iod_10-brom_11-brom_13-chlor_15-chlor.log (2.3 SDs)
- 6Heliceno_7-chlor_9-chlor_15-fluor_16-fluor.log (3.1 SDs)
- 6Heliceno_3-iod_7-fluor_8-brom_9-brom_15-brom.log (3.3 SDs)
- 6Heliceno_2-chlor_3-brom_7-iod_9-chlor_14-fluor_15-iod.log (3.2 SDs)
- 6Heliceno_8-fluor_11-fluor_12-chlor_14-chlor_16-iod.log (3.1 SDs)
- 6Heliceno_1-fluor_2-iod_3-brom_4-iod.log (4.5 SDs)

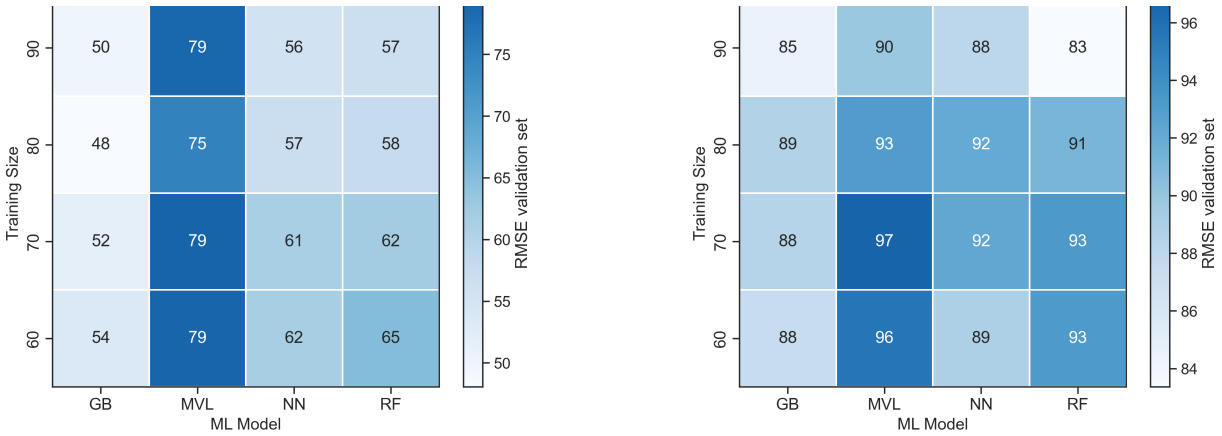
Test: 49 outliers out of 235 datapoints (20.9%)
- 6Heliceno_1-fluor_4-brom_5-brom_9-iod_11-fluor_16-brom.log (3.9 SDs)
- 6Heliceno_2-fluor_5-fluor_8-chlor_12-iod_13-brom_16-fluor.log (4.9 SDs)
- 6Heliceno_12-chlor_13-iod_16-fluor.log (2.1 SDs)
- 6Heliceno_4-chlor_6-iod_8-chlor_12-iod_13-iod_16-fluor.log (2.1 SDs)
- 6Heliceno_4-fluor_5-fluor_6-iod_7-iod_10-brom_13-fluor.log (4.2 SDs)
- 6Heliceno_10-iod_11-iod_14-iod.log (2.9 SDs)
- 6Heliceno_6-iod_9-brom_10-chlor_11-iod_12-brom_14-brom.log (2.3 SDs)
- 6Heliceno_6-fluor_8-chlor_11-chlor_13-fluor_15-iod_16-iod.log (7.9 SDs)
- 6Heliceno_1-fluor_4-brom_5-fluor_6-fluor_8-fluor_10-iod.log (2.9 SDs)
- 6Heliceno_1-brom_3-fluor_10-iod_11-iod_14-chlor.log (6.1 SDs)


SD of the errors
SD of the errors
10
5
0
−5
−10
−10
−5
0
5
10


SD of the errors
SD of the errors
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

Section F. Model Screening

This section compares different combinations of hyperoptimized algorithms and partition sizes.



Section G. Reproducibility

This section provides all the instructions to reproduce the results presented.

1. Download these files *(the authors should have uploaded the files as supporting information!)*:

- CSV database (reflected_dataset_Rmax.csv)

2. Install and adjust the versions of the following Python modules:

- Install ROBERT and its dependencies: conda install -y -c conda-forge robert
- Adjust ROBERT version: pip install robert==1.2.1
- Install scikit-learn-intelex: pip install scikit-learn-intelex==2025.0.1

(if scikit-learn-intelex is not installed, slightly different results might be obtained)

3. Run ROBERT using this command line in the folder with the CSV database:

python -m robert --ignore "[lambd,Nsubs,Names]" --names "Names" --y "R" --csv_name "reflected_dataset_Rmax.csv" --corr_filter "False" --shap_show "16"

4. Execution time, Python version and OS:

Originally run in Python 3.10.16 using Windows 10.0.26100

Total execution time: 477.48 seconds *(the number of processors should be specified by the user)*

Section H. Transparency

This section contains important parameters used in scikit-learn models and ROBERT.

1. Parameters of the scikit-learn models (same keywords as used in scikit-learn):

No PFI (standard descriptor filter):	**PFI (only most important descriptors):**
sklearn model: GradientBoostingRegressor	sklearn model: RandomForestRegressor
random_state: 19	random_state: 43
names: Names	names: Names
n_estimators: 100	n_estimators: 60
max_depth: 5	max_depth: 100
max_features: 0.5	max_features: 0.75
min_samples_split: 2	min_samples_split: 2
min_samples_leaf: 1	min_samples_leaf: 1
min_weight_fraction_leaf: 0	min_weight_fraction_leaf: 0
ccp_alpha: 0	ccp_alpha: 0
learning_rate: 0.1	oob_score: False
subsample: 1.0	max_samples: 0.75
validation_fraction: 0.1	

2. ROBERT options for data split (KN or RND), predict type (REG or CLAS) and hyperopt error (RMSE, etc.):

No PFI (standard descriptor filter):	**PFI (only most important descriptors):**
split: RND	split: RND
type: reg	type: reg
error_type: rmse	error_type: rmse

Section I. Abbreviations

Reference section for the abbreviations used.

ACC: accuracy
ADAB: AdaBoost
CSV: comma separated values
CLAS: classification
CV: cross-validation
F1 score: balanced F-score
GB: gradient boosting
GP: gaussian process
KN: k-nearest neighbors
MAE: root-mean-square error
MCC: Matthew's correl. coefficient
ML: machine learning
MVL: multivariate lineal models
NN: neural network
PFI: permutation feature importance
R2: coefficient of determination
REG: Regression
RF: random forest
RMSE: root mean square error
RND: random
SHAP: Shapley additive explanations
VR: voting regressor

Miscellaneous

General tips to improve the models and instructions to predict new values.

Some general tips to improve the score

1. Adding meaningful datapoints might help to improve the model. Also, using a uniform population of datapoints across the whole range of y values usually helps to obtain reliable predictions across the whole range. More information about the range of y values used is available in Section C.

2. Adding meaningful descriptors or replacing/deleting the least useful descriptors used might help. Feature importances are gathered in Section D.

How to predict new values with these models?

1. Create a CSV database with the new points, including the necessary descriptors.

2. Place the CSV file in the parent folder (i.e., where the module folders were created)

3. Run the PREDICT module as 'python -m robert --predict --csv_test FILENAME.csv'.

4. The predictions will be shown at the end of the resulting PDF report and will be stored in the last column of two CSV files called MODEL_SIZE_test(_No)_PFI.csv, which are in the PREDICT folder.
